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Study on the thermodynamics of the gadolinium-hydrogen binary system (H/Gd = 0.0–2.0) and implications to metallic gadolinium purification

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ABSTRACT

The thermodynamics of the gadolinium-hydrogen (Gd–H) binary system (H/Gd = 0.0–2.0) between 650 and 900 °C is studied by pressure composition isotherm measurement. Significant H dissolution in Gd is observed, up to H/Gd = 0.34 at 650 °C and 0.55 at 900 °C. The metal-rich phase boundary of non-stoichiometric gadolinium dihydride was found to occur with nominal composition of $GdH_{1.80}$ at 650 °C and $GdH_{1.53}$ at 900 °C. The results are in fairly good agreement with previous experimental work but with improved accuracy. The binary phase diagram is obtained using the CALPHAD method. The thermodynamic study here assists the understanding on the efficient deoxygenation effect by the H in Gd for Gd purification.

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1. Introduction

The interaction of rare earth (RE) and hydrogen has long been the subject of many investigations owing to their exceptional physical and structural properties that can be tuned by changing their hydrogen content. Rare earth-based hydrogen storage alloys have received an increasing amount of attention [1–5]. Rare earth metal hydrides can serve as smart coatings which can switch their optical properties reversibly between reflective and transparent states by hydrogenation and dehydrogenation [6–9]. In addition, our recent experimental results showed that oxygen in gadolinium metal could be removed effectively by thermal treatment of Gd with controlled amount of hydrogen in Gd in vacuum [10]. Thus, it is essential to have a fundamental knowledge about the RE-H systems, such as information about the phase equilibrium and the related thermodynamic data.

The thermodynamic properties of the La–H [11,12], Ce–H [13] Sm–H [14,15] and Er–H [16] systems have been well studied by various experimental and computational techniques. Phase diagrams with sufficient accuracy have been established. However, the interaction of hydrogen with the heavier rare earth metals (e.g. Gd–H, Tb–H and Dy–H systems) are less studied [17]. In this paper, we report a detailed measurement of the Gd–H pressure composition isotherms (PCI) in the temperature range 650–900 °C and establish a Gd–H phase diagram using the newly measured PCI data. Our study focuses on the hydrogen concentration region with H/Gd = 0.0–2.0, i.e. the Gd–H solid solution and β-GdH_2 region, which is closely related to our recent finding on the deoxygenation in Gd metal by the dissolved hydrogen [10].

Data related to the Gd–H phase diagram from current literature is summarized in Table 1. Previous PCI measurements [18–22], provided a general picture of that Gd–H binary system, which shows a large solubility of hydrogen in α-Gd, and two non-stoichiometric hydrides (β-GdH_2 and γ-GdH_3). In addition to the PCI measurements, Beaudry has carried out several measurements on the Gd and the two phase region (α-Gd +β-GdH_2) using an equilibration analytical technique between 250 °C and 850 °C [23]. Beaudry also made some XRD measurement, which indicates a

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Table 1
List of the experimental data of the literature.

Composition range	Temperature range (°C)	Experimental technique	Reference	Data used for modeling
α-Gd-GdH_2	250 < T < 850	Equilibration analysis	[23]	Yes
	500	X-ray-diffraction	[25]	Yes
	650 < T < 900	PCI	This work	Yes
	700 < T < 775	PCI	[22]	Yes
	600 < T < 800	PCI	[18]	No
	700 < T < 1000	PCI	[19]	No
	800 < T < 900	PCI	[20]	No
	650 < T < 1050	PCI	[21]	No
GdH_2-GdH_3	325	TG-DTA	[24]	Yes
	200 < T < 300	PCI	[18]	Yes

quiet low solubility (<0.1 at.%) of H in Gd at ambient temperature. A TG-DTA measurement on the GdH_2-GdH_3 system was carried out by Mintz [24].

Unfortunately, the reported information is not always very consistent with each other, which will impede the effective usage of such information. PCI measurements remain the most accurate and informative technique to study the Gd–H phase diagram, which is also the most time consuming. In most PCI measurements, the initial and terminal points of the plateaus corresponding to the transition from α-Gd to β-GdH_2, which determine the phase boundaries in the phase diagram, were evaluated without sufficient accuracy. Clearly, PCI measurements on the Gd–H system with higher accuracy, particularly for the points determining the phase boundaries, are highly desirable to provide more quantitative understanding on the fundamental thermodynamics of this system and assist the new applications related to the H in the rare earth metals.

2. Experimental

The high purity Gd containing gaseous impurities less than 50 ppm was used in this work. Commercial Gd (above 99.5 mass% in purity by the supplier) was refined by hydrogen plasma arc melting (HPAM) using a laboratory-scale plasma arc furnace under Ar–H_2 (20 vol.%) atmosphere. The detailed information about the operation procedure were reported in our previous work [26]. Metallic and gaseous impurities concentrations were precise analyzed by inductively coupled plasma-atomic emission spectrometer (ICP-AES) and inert gas fusion analysis (IGA), respectively. Impurity concentrations in Gd metal after plasma arc melting are shown in Table S1 and the final purity of Gd increases to above 99.95%. The GdH_3 powders were prepared using a stainless steel vacuum tubular furnace by hydriding the high purity metallic gadolinium with an original hydrogen pressure of 4 MPa at 300 °C for 12 h and slowly cooling down to room temperature under hydrogen pressure. Fig. S1 shows the XRD patterns of the as-prepared powders, indicating the complete transformation to the GdH_3 phase (ICSD 77970).

The PCI measurement is carried out in a home designed Sievert apparatus using high purity Gd. The whole system can be pumped down to a base pressure of about 8×10^{-8} Torr (~1.1×10^{-5} Pa) using a turbomolecular pump, with a low leaking rate of 8.4×10^{-4} Pa cm^3 s^{-1}. Two capacitance pressure gauges with ranges from 10^{-4} to 1 Torr and 10^{-1} to 10^3 Torr respectively, were used to monitor the pressure in the system with sufficient sensitivity and accuracy. The lowest limit of the pressure, measured by the 1 Torr pressure gauge, was about 1×10^{-4} Torr, but for accuracy reasons we used measurements above 10^{-3} Torr only. The amount of hydrogen was then calculated by a volumetric method in which a resolution of 9×10^{-6} mol of hydrogen in the specimen could be obtained. The data from the pressure gauges was acquired automatically by a computer controlled data acquisition system. A controlled resistance furnace was used to heat the reactor and the temperature was controlled using a PID controller. The temperature was measured with a Chromel-Alumel thermocouple with precision of ±0.1 °C on the outside of the reactor.

The PCI measurement was carried out at five different temperature between 650 and 900 °C. The Gd sample is an ingot ~0.5 g, which allows determining the H/M value with sufficient accuracy. For each hydrogen dosing step, the equilibrium hydrogen pressure was measured after it had remained constant for at least 0.5 h. The hydrogen dosing was manually controlled using a needle valve and elaborately controlled so that the interval between data points around the phase boundaries are less than 0.05 H/M. Large number of experimental points were obtained especially at each end of the plateaus. In order to avoid possible cumulative oxygen contamination, a new gadolinium sample was used for each measurement.

Differential scanning calorimetry (DSC) measurement on a GdH_3 sample was performed on a NETSCH DSC 204 HP calorimeter to examine the phase transition temperatures. About 5 mg as-prepared GdH_3 powders was introduced in an aluminum pan and the measuring cell was continuously purged with high purity hydrogen gas at a flow rate of 50 mL min^{-1}. The temperature of phase transition was determined under 1 and 10 MPa H_2 respectively with the heating rate of 2 °C/min.

Finally, the Gd–H binary phase diagram has been assessed with the CALPHAD method using the Pandat phase equilibrium calculation software.

3. Results and discussion

3.1. Characterization of Gd–H binary system

3.1.1. Pressure-composition-isotherms

The hydrogenation isotherms are measured at six different temperatures from 650 to 900 °C, as illustrated in Fig. 1. One well defined plateau between H/Gd from 0.3 to 1.8 is observed for each temperature, which is corresponding to be the GdH_2/Gd biphasic region. The width of the plateau gradual shrinks with increasing temperature. The above features are characteristic of a typical hydrogen induced first order phase change behavior. The hydrogen partial pressure for the plateaus is in very good agreement with previous results [18–22]. The strong dependence of the hydrogen partial pressure with the H/Gd ratio in the low H/Gd region suggests substantial H solubility in metallic Gd.

In the literature [18], the number of the measurement points was not sufficient, thus the initial and terminal points of the plateaus corresponding to the transition from α-Gd to β-GdH_2 were evaluated without sufficient accuracy. For example, The terminal hydrogen solubility in Gd, which is equal to the H/M concentration for which the plateau pressure starts, can be only ascertain in the range of H/M = 0.334–0.572 at 650 °C and H/M = 0.425–0.556 at

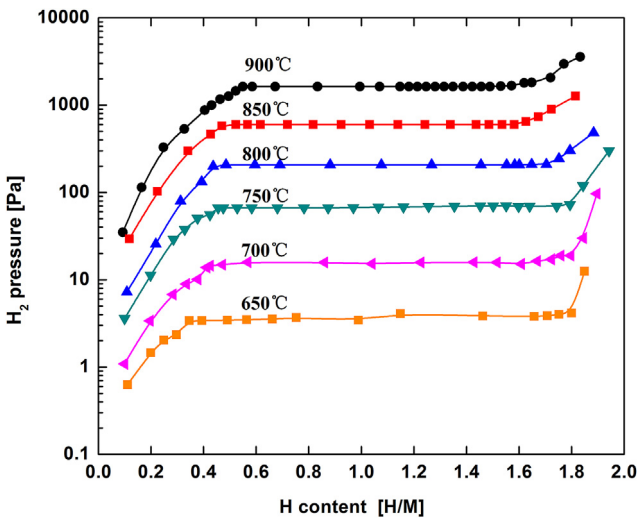

Fig. 1. Pressure-composition isotherms measured from 650 °C to 900 °C in H/Gd = 0.0–2.0 (α-Gd and β-GdH_2).

800 °C. The metal-rich phase boundary of nonstoichiometric gadolinium dihydride was found to occur in the range of H/M = 1.817–1.932 at 650 °C and H/M = 1.590–1.811 at 800 °C. These experimental data cannot be used to deduce phase boundary accurately and should be rejected from the optimization in CALPHAD technique. One of the good example that can be cited is that the Nd–H phase diagram calculated by Luo et al. [27] is deviate from PCI data reported by Mulford [28] because the initial and terminal points of the plateaus were evaluated without sufficient accuracy. And these PCI data were not used for the optimization of the Nd–H system to ensure the accuracy of the calculated phase diagram.

High quality PCI curves of Gd–H system have been measured by Beneznitsky [22] to obtain the terminal hydrogen solubility in Gd. But the absorption isotherms were measured in the narrow range of temperature (700–775 °C) and composition (H/M < 0.5). The limited experimental data is insufficient to accomplish the CALPHAD technique.

In our measurement, The PCI curves of Gd–H system were measured between the Gd solid solution α-Gd and the dihydride β-GdH_2 in the temperature range 650–900 °C. The hydrogen dosing was manually controlled using a needle valve and elaborately controlled so that the start and end points of each plateau are more accurately determined compared to previous studies, which allows more accurate determining the phase boundaries. The interval between data points around the phase boundaries are controlled less than 0.05 H/M. Fig. S2 presents the phase boundaries deduced from the above PCI curves compared to the experimental values taken from literature [18,22]. The phase boundaries determined in this study are in fairly good agreement with previous experimental work but with improved accuracy.

3.1.2. Thermodynamic properties of phase transformation

The flat plateau in the PCI curves is characteristic of a typical first order phase transition. The plateau pressures P_{eq} is corresponding to the hydrogen partial pressure for the α-β phases at the corresponding temperature. P_{eq} is associated with the fundamental thermodynamic parameters through the well-known van't Hoff equation for the temperature dependence of the plateau pressures:

$$\ln P_{eq} = \frac{\Delta H_f^0}{RT} + \frac{\Delta S_f^0}{R} \tag{1}$$

where ΔH_f^0 and ΔS_f^0 are the enthalpy and entropy of the hydride formation, respectively. The $\ln P_{eq}$ vs. 1/T should be a straight line, which is indeed observed, as shown in Fig. 2. The excellent linearity is evident of the high accuracy of the PCI measurement. From the slope, we can obtain the formation enthalpy of the β-GdH_2 with high accuracy $\Delta H_f^0 = (-208.5\pm1.2)$ kJ/mol H_2. The PCI measurements by Sturdy et al. [18] gave $\Delta H_f^0 = (-196.6\pm7.1)$ kJ/mol H_2, which is slightly higher than our result. However, if the 600 °C data point which was clearly deviated from data obtained at higher temperature (Fig. 2) was excluded from Sturdy's analysis, we obtain $\Delta H_f^0 = (-210.0\pm1.9)$ kJ/mole H_2, and this result is in excellent agreement with our measurement. We attribute the deviations in Sturdy's study to slow kinetics of hydrogen absorption under 600 °C.

3.2. Thermodynamic assessment of the binary Gd–H system

3.2.1. The empirical Gibbs free energy model of the Gd–H binary system

With the experimental thermodynamic and phase boundary data in this work, the Gd–H binary phase diagram can be assessed with the Calculation of Phase Diagrams (CALPHAD) method. In the CALPHAD method, the Gibbs free energy of each individual phase is modeled and the model parameters are optimized by fitting all available piecewise experimental (or theoretical) thermodynamic information.

Specifically, for the Gd–H binary system, the Gibbs energy functions of the pure elements (Gd and H) are taken from the SGTE (Scientific Group Thermo data Europe) unary database [29], expressed by the following equation:

$$^0G(T) = a + bT + cT\ln(T) + dT^2 + eT^3 + \frac{f}{T} + \sum_n g_n T^n \tag{2}$$

where a to f and g_n are documented coefficients.

The di-hydrogen H_2 is the only specie considered in the gas

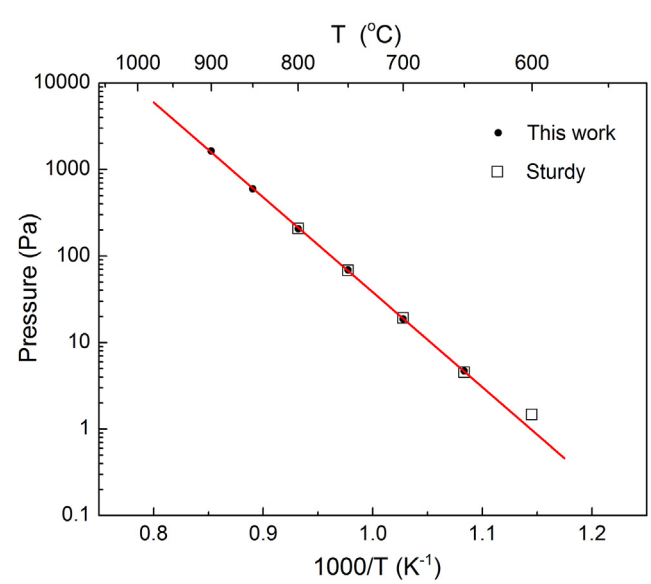

Fig. 2. Van't Hoff plots correspond to the metal to hydride phase transition compared to Sturdy's measurements.

phase and the Gibbs free energy was given by:

$$G^{gas} = {}^{ref}G^{gas}_{H_2} + RTln(P_{H_2}) \quad (3)$$

Below 1235 °C, the packing of Gd atoms is hexagonal closest packing (hcp) for metallic Gd and γ-GdH_3 while is face centered cubic (fcc) for β-GdH_2. Incorporation of atomic H in Gd metal can be regarded as filling the H into the tetrahedral or octahedral interstitial sites. In both fcc and hcp structure, there are one relatively larger yet more weakly absorbing interstitial octahedral (O) site and two more energetically favorable interstitial tetrahedral (T) sites per Gd atom [30,31]. Therefore, a three sublattice model [32] is used to simplify the model for the Gd–H solid phases. The three sublattices is given by $(Gd)(H,Va)^O(H,Va)^T_2$, composed of one sublattice fully occupied by the Gd atoms and two other sublattices corresponding to the octahedral (O) and tetrahedral (T) interstitial sites. The term $(H,Va)^O$ means that the O sublattices can be either occupied by a H atom (H) or leaving unoccupied (Va), which is similar for the T sublattice. The same thermodynamic model are used for α-Gd and γ-GdH_3 because their crystal structure are similar. If the hydrogen atoms occupied the same interstices with same fraction, the Gibbs free energy of formation should be same. So both the α-Gd and γ-GdH_3 are described as a single hcp phase, consistently with their crystal structures.

Gadolinium has a hcp structure at room temperature and transforms to a body-centered cubic (bcc) structure at 1235 °C, very close to its melting point (1313 °C). It is difficult to study the bcc phase region at such a high temperature under hydrogen pressure. To the best of our knowledge, interstitial locations of hydrogen, phase equilibria and thermodynamic data focusing on the bcc phase is absent currently.

In this work, bcc phase is considered to dissolve hydrogen interstitially, described by a two sublattice model $(Gd)(H,Va)_u$ with the first sublattice fully occupied by Gd and the second sublattice represents the interstitial sites occupied by hydrogen atoms and vacancies. If it were assumed to occupy the T-site (tetrahedral) in the bcc form, the value of u should be 6. Konisberger [33] discussed the large discrepancy between the geometric value and the actual u used for bcc form. The value of u for bcc is generally smaller in order to avoid more model parameters for describing the large excess Gibbs energies and site blocking effect. In present work, the value of u is assumed to 1, as same as the model used in Sm–H system for the bcc phase [15].

The Gibbs free energy of a given solid Gd–H phase is given by

$$G^{\varphi} = {}^{ref}G^{\varphi} + {}^{id}G^{\varphi} + {}^{ex}G^{\varphi} \quad (4)$$

Where ${}^{ref}G^{\varphi}$, ${}^{id}G^{\varphi}$ and ${}^{ex}G^{\varphi}$ are the reference, ideal and excess Gibbs free energy.

For φ = hcp and fcc, where two interstitial lattices represent tetrahedral and octahedral sites:

$$\begin{aligned} {}^{ref}G^{\varphi} = {} & y'_{Va}y''_{Va}G^{\varphi}_{Gd:Va:Va} + y'_{H}y''_{Va}G^{\varphi}_{Gd:H:Va} + y'_{Va}y''_{H}G^{\varphi}_{Gd:Va:H} \\ & + y'_{H}y''_{H}G^{\varphi}_{Gd:H:H} \end{aligned} \quad (5)$$

$${}^{id}G^{\varphi} = RT\left(\left(y'_{H}lny'_{H} + y'_{Va}lny'_{Va}\right) + 2\left(y''_{H}lny''_{H} + y''_{Va}lny''_{Va}\right)\right) \quad (6)$$

$$\begin{aligned} {}^{ex}G^{\varphi} = {} & y'_{H}y'_{Va}y''_{Va}L^{\varphi}_{Gd:H,Va:Va} + y'_{H}y'_{Va}y''_{H}L^{\varphi}_{Gd:H,Va:H} \\ & + y'_{H}y''_{H}y''_{Va}L^{\varphi}_{Gd:H:H,Va} + y'_{Va}y''_{H}y''_{Va}L^{\varphi}_{Gd:Va:H,Va} \end{aligned} \quad (7)$$

y'_H and y'_{Va} are the fraction of the H occupied and vacant second sublattice sites and y'' refers to site fraction in the third. The energy terms $G^{\varphi}_{Gd:O:T}$, where O and T could be either H or Va, describes the Gibbs free energy of a certain H occupancy type of the three sublattices which has four possibilities. The energy term L^{φ} describes the binary interaction energy which is temperature dependent.

For φ = bcc, where only one interstitial lattice is present:

$${}^{ref}G^{\varphi} = y'_{Va}G^{\varphi}_{Gd:Va} + y'_{H}G^{\varphi}_{Gd:H} \quad (8)$$

$${}^{id}G^{\varphi} = RT\left(y'_{H}lny'_{H} + y'_{Va}lny'_{Va}\right) \quad (9)$$

$${}^{ex}G^{\varphi} = y'_{H}y'_{Va}L^{\varphi}_{Gd:H,Va} \quad (10)$$

where y'_H and y'_{Va} are the fraction of the H occupied and vacant second sublattice sites. $G^{\varphi}_{Gd:Va}$ and $G^{\varphi}_{Gd:H}$ are the expressions for the Gibbs energy of the pure bcc Gd and the hypothetical compounds GdH_1 with bcc arrangements of gadolinium atoms, respectively.

The binary interaction parameters are described as linear functions of temperature

$$L^{\varphi} = A + BT \quad (11)$$

where T is the temperature in absolute scale and A, B are fitting parameters.

The liquid phase has not been considered due to the absence of any experimental data.

3.2.2. Thermodynamic optimization

The Pan-optimizer module of the Pandat software is used for thermodynamic optimization. Our measurement results and some of the experimental data in Table 1 are used in the fitting. The data showing apparent inconsistency with the most data sets or only evaluated roughly without accurate measurements are not included in the optimization. The experimental phase equilibria data [18,22,23,25] and thermochemical data [24] are assigned a high weight for the optimization. During an initial guess with the fitting parameters, the entropy of formation for the atomic H is 65 J mol^{-1} K^{-1}. The thermodynamic parameters for fcc and hcp phases are optimized at the first stage using the phase boundaries information in Table 1. Then, the optimization is continued by taking into account bcc phase. Because of the lack of any experimental data, For the enthalpy of formation of the end-member GdH (bcc) as a first approximation, the value of the GdH fcc phase has been used, i.e. ΔH_{GdH} $(bcc) = -46654$ J·mol^{-1}. All the parameters were finally evaluated together to give the best description of the system.

The calculated phase diagram is reported in Fig. 3 and the obtained parameters are listed in Table 2. The calculated phase boundaries in this work is in excellent consistency with our new reported experimental data in the temperature region 650–900 °C. It also fairly well reproduces the experimental phase boundaries from most of the previous works. In the PCI measurement by Sturdy et al. [18], the initial and ending points in the PCI plateau are obtained by extrapolation. The corresponding phase boundaries are also in agreement with our calculated ones. Two invariant reactions can be found in this evaluation are listed in Table 3.

In order to check the quality of the modeling, PCI curves are simulated using the obtained parameters in Table 2. As shown in Fig. 4, the calculated PCI curves are in very good agreement with our experimental measurements for the GdH_2/α–Gd transition.

The GdH_2/GdH_3 boundaries are obtained solely based on the previous experimental data. To check the quality of modeling in this region, differential scanning calorimetry (DSC) measurements on a GdH_3 sample in an hydrogen gas flow (50 mL min^{-1}, 1.0 atm and 10 atm) is carried out. Under hydrogen pressure of 1 atm (Fig. 5a), a

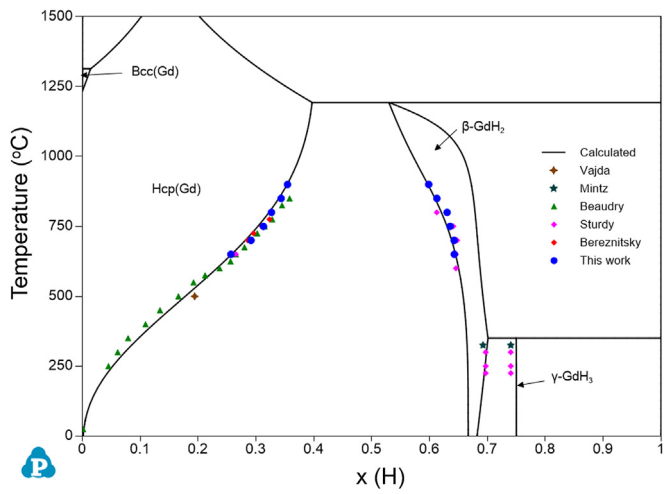

Fig. 3. The calculated Gd–H binary phase diagram compared with some experimental measurements used during the modeling.

Table 2
The optimized parameters of the binary Gd–H system.

Phase	Parameters
hcp	$G^{Hcp}_{Gd:Va:Va} = G^{SER}_{Gd}$
	$G^{Hcp}_{Gd:H:Va} = G^{SER}_{Gd} + G^{SER}_{H} - 77975.16 + 65 \times T$
	$G^{Hcp}_{Gd:Va:H} = G^{SER}_{Gd} + 2 \times G^{SER}_{H} - 143204.66 + 130 \times T$
	$G^{Hcp}_{Gd:H:H} = G^{SER}_{Gd} + 3 \times G^{SER}_{H} - 254501 + 209.77 \times T$
	${}^{0}L_{Gd:*:H,Va} = -29398.22 + 4.2503 \times T$
fcc	$G^{Fcc}_{Gd:Va:Va} = G^{SER}_{Gd} + 6232 - 1.6835667 \times T$
	$G^{Fcc}_{Gd:H:Va} = G^{SER}_{Gd} + G^{SER}_{H} - 46654 + 65 \times T$
	$G^{Fcc}_{Gd:Va:H} = G^{SER}_{Gd} + 2 \times G^{SER}_{H} - 208455.53 + 143.28 \times T$
	$G^{Fcc}_{Gd:H:H} = G^{SER}_{Gd} + 3 \times G^{SER}_{H} - 230664.25 + 195 \times T$
	${}^{0}L_{Gd:*:H,Va} = 55779.89 - 22.6207 \times T$
	${}^{0}L_{Gd:H,Va:*} = -21032.178$
bcc	$G^{Bcc}_{Gd:Va} = G^{Bcc}_{Gd}$
	$G^{Bcc}_{Gd:H} = G^{Bcc}_{Gd} + G^{SER}_{H} - 46654 + 65 \times T$

Table 3
Invariant reactions in the Gd–H system at 1 atm hydrogen pressure.

Reaction	T (°C)	Composition (mol% H)		
β-GdH_2+Gas(H_2)→γ-GdH_3	350.2	0.701	1.000	0.750
Hcp(Gd) +Gas(H_2)→β-GdH_2	1193.6	0.397	1.000	0.530

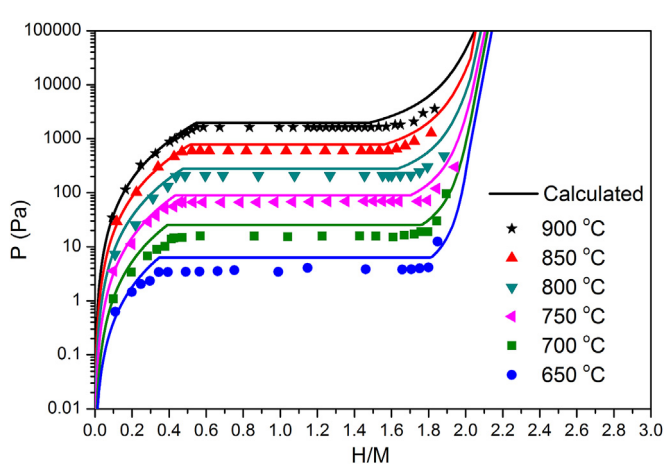

Fig. 4. Calculated PCI curves compared with experimental measurements.

clearly endothermic peak is observed at about 354.6 °C, which is corresponding to the transition $GdH_3 \rightarrow GdH_2 + H_2(gas)$. This experimental result is very close to the 350.2 °C gained by CALPHAD method. While under 10 atm (Fig. 5b), the calculated decomposition temperature increase to 431.4 °C due to the increasing stability of GdH_3. And this result is verified experimentally by DSC measurement with a endothermic peak appears at 431.3 °C.

3.3. Implication of the deoxygenation of Gd using dissolved H

In our recent work [10], we demonstrate a novel refining method to prepare high purity Gd metal by deoxygenation using the dissolved H. Due to the strong oxygen affinity of the lanthanide metals, oxygen is the most persistent non-metal impurity in Gd and also other lanthanide metals. Directly deoxygenation using H_2 (reaction (12)), however, is thermodynamically forbidden within reasonable temperature due to the strong oxygen affinity of Gd [34].

$$H_2(g) + O(in\ Gd) = H_2O(g) \tag{12}$$

Atomic H is a stronger reducing agent. We show that using hydrogen thermal plasma treatment of the melted Gd, effective deoxygenation has been achieved [35]. The key species is the highly active atomic H generated by plasma. The thermal plasma approach, however, has the drawback of high processing temperature (up to 10^4 K), which will cause significant evaporation of the Gd metal. Using the H dissolved in Gd metal is an effective alternative to generate active H atoms in moderate temperature. The deoxygenation is through the following reaction:

$$2H(in\ Gd) + O(in\ Gd) = H_2O(g) \tag{13}$$

A negative value for ΔG_{13} means the thermodynamics feasibility of reaction (13), i.e., the deoxidation of gadolinium by dissolved H atom are estimated to occur spontaneously. The typical deoxygenation reaction is carried out at 1400 K in a high vacuum furnace at background pressure of 10^{-5} Pa. At this condition, the Gibbs free energy change of reaction (13) (ΔG_{13}) is plotted in Fig. 6, against the H concentration in Gd. During the calculation the standard Gibbs energy of H (in Gd) related to the hydrogen content is obtained by thermodynamic assessment of Gd–H system in this work, and other required data are extrapolated based on the results of [34,36]. ΔG_{13} rapidly decreases with the H/Gd ratio at very low H concentration and the decrease becomes much slower for H concentration higher than 400 ppm. The substantial negative value of ΔG_{13} (-48.7 kJ mol^{-1} H^{-1} at H concentration of 400 ppm) suggests that the hydrogen atoms dissociated in Gd metal can actively react with oxygen impurities to form water. The formed H_2O molecules are readily removed at elevated temperatures, which was proved in our previous study [10]. The interaction of water vapor with gadolinium surface cannot form stable gadolinium hydroxide when temperature is higher than 500 °C [37]. Meanwhile, the dissolved hydrogen can be easily recombined into gaseous hydrogen and leave the metal without making any recontamination.

The difference between the two reactions is the recombination of the dissolved H in Gd into H_2 molecules:

$$2H(in\ Gd) = H_2(g) \tag{14}$$

That is, the difference between Gibbs energy change for reaction (12) and reaction (13) can be determined from the value of $\Delta G_{14} = \Delta G_{12} - \Delta G_{13}$. A negative ΔG_{14} means the H in Gd is less stable, i.e., possesses a higher reduction potential for deoxygenation. The Gibbs free energy change of reaction (14) (ΔG_{14}) is

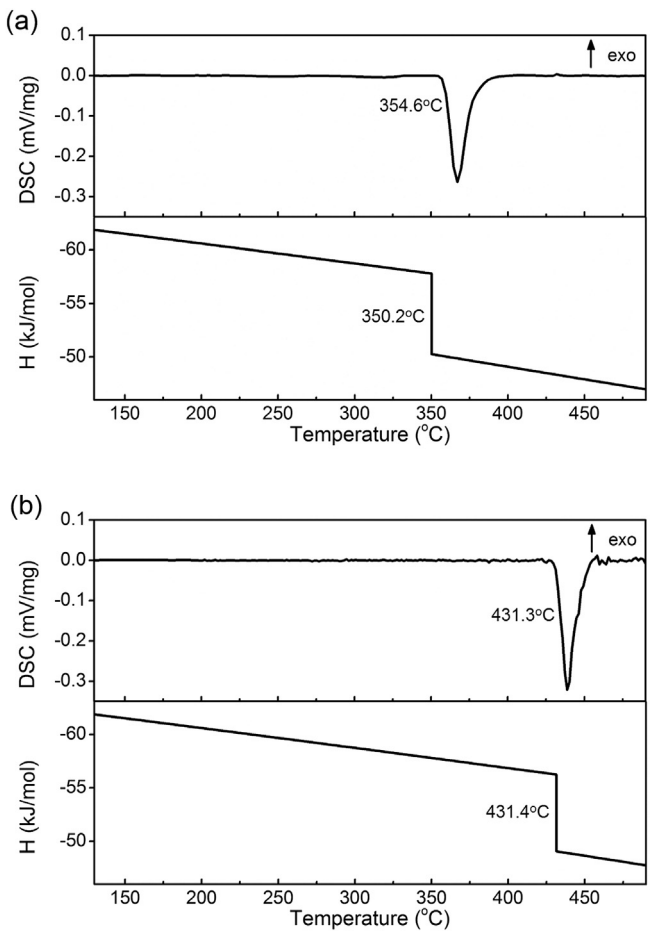

Fig. 5. DSC curve and calculated enthalpy value of GdH_3–GdH_2 system at (a) 1 atm and (b) 10 atm hydrogen pressure.

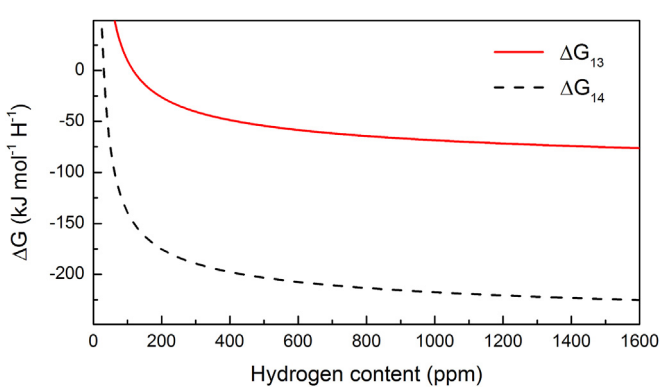

Fig. 6. Gibbs free energy change for the reaction (13) and (14) as a function of the content of dissolved hydrogen in Gd.

plotted in Fig. 6, against the H concentration in Gd. The substantial negative value of ΔG_{14} means a large free energy gain of reaction (13) compared to reaction (12), suggesting that the deoxygenation by dissolved H in Gd is thermodynamically more favorable compared to using hydrogen gas.

The experimental and theoretical results obtained here also provide insights into the proper processing condition for the deoxygenation. Although higher H concentration is thermodynamically more favorable, the marginal Gibbs free energy gain gradually diminishes with the increasing H concentration. On the other hand, formation of the GdH_2 phase will cause pulverization of the sample, which significantly enlarges the surface area and increases the risk of oxygen contamination when exposed to air compared to an integrated ingot. Therefore, the H_2 partial pressure during the hydrogenation should be carefully controlled so that the sample remains in the α phase region.

4. Conclusions

A combined study by PCI measurement and phase diagram calculation is carried out to provide more quantitative understanding on the thermodynamics of the gadolinium-hydrogen (Gd–H) binary system (H/Gd = 0.0–2.0) between 650 and 900 °C. The phase boundaries of the two phase region is determined with high accuracy, which is from H/Gd = 0.34 to 1.80 at 650 °C and 0.55 to 1.53 at 900 °C. The calculated phase diagram is also in good agreement with previous experimental studies. The thermodynamic evaluation shows that the H in the metallic Gd is thermodynamically more favorable for the deoxygenation reaction compared to H_2 gas, which provides an effective method for purification of Gd metal.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jallcom.2016.02.201.

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